

Introduction

BWA-MEM and Bowtie2 are the dominant short-read DNA aligners in modern sequencing pipelines. Both use a Burrows-Wheeler-transformed (BWT) index of the reference genome to achieve rapid exact and approximate matching. The BWT compresses the reference while preserving the ability to query for substrings efficiently — the algorithmic insight that brought genome-scale alignment from hours per sample to minutes. Production alignment runs at the command line; R wrappers (**Rbowtie2**, **BWA** via **Rsubread**) integrate with downstream Bioconductor analyses.

Prerequisites

A working understanding of FASTQ format, reference genomes, and the SAM/BAM alignment format.

Theory

BWA-MEM (Burrows-Wheeler Aligner, Maximal Exact Matches) seeds local alignments by finding maximal exact matches, then extends via banded Smith-Waterman. It is the default aligner for whole-genome DNA sequencing because of its accuracy on long-insert paired-end reads and its handling of indels.

Bowtie2 offers end-to-end alignment (the read must align contiguously) or local alignment (soft-clipping allowed at read ends). It is fast and accurate for short reads and is a common choice for ChIP-seq, ATAC-seq, and small-genome DNA sequencing.

Both write SAM (text) or BAM (binary) output, including alignment scores, mapping quality, mate-pair information, and tags for soft-clipping, mismatches, and other features. For long reads (PacBio, Nanopore), use minimap2 instead — it is designed for the higher error rates and longer alignment lengths.

Assumptions

Short reads (typically below ~300 bp; for longer, switch to minimap2); reference genome appropriate to the species and assembly version of the experiment.

R Implementation

```
library(Rbowtie2); library(Rsamtools)

# Build index (one-off)
# bowtie2_build(references = "reference.fa", bt2Index = "idx")

# Align
# bowtie2(bt2Index = "idx", samOutput = "aln.sam",
#         seq1 = "reads.fastq", overwrite = TRUE)

# Convert SAM to sorted, indexed BAM
# asBam("aln.sam", destination = "aln_sorted")
# indexBam("aln_sorted.bam")

# Read a short segment
# scanBam("aln_sorted.bam", param = ScanBamParam(which = GRanges("chr1", IRanges(1, 1000))))
```

Output & Results

The primary output is a BAM file containing aligned reads with mapping quality, CIGAR string, and tags. **samtools flagstat** (or **Rsamtools::quickBamFlagSummary()**) reports alignment statistics: total reads, reads aligned, properly paired, duplicates, secondary alignments. A 95–98 % alignment rate is typical for high-quality WGS data.

Interpretation

A reporting sentence: “BWA-MEM (v0.7.17) alignment to GRCh38 achieved 97 % overall alignment rate on a 30× whole-genome library, with <1 % ambiguously mapped reads (multi-mappers); after Picard MarkDuplicates, the duplicate rate was 8 %, consistent with a well-optimised PCR-based library prep.” Always state the aligner version, reference assembly, and alignment / duplicate rates.

Practical Tips

- Use minimap2 for long reads (PacBio HiFi, Oxford Nanopore); BWA-MEM and Bowtie2 are short-read tools and lose accuracy beyond about 300 bp.
- Report alignment statistics (**flagstat**) after alignment as a routine QC step; declines in alignment rate signal contamination, library-prep problems, or reference mismatch.
- Index BAM files (**samtools index** or **Rsamtools::indexBam()**) for random-access by downstream tools (variant calling, peak calling, browser visualisation).
- Deduplicate after alignment with Picard MarkDuplicates or **samtools markdup**; PCR duplicates inflate variant-calling false positives if not removed.
- For paired-end reads, specify both mate files; the aligner uses insert-size constraints to improve mapping accuracy near repeats.
- For ChIP-seq and ATAC-seq, Bowtie2 with **--very-sensitive** is the standard; for WGS variant calling, BWA-MEM is preferred.

R Packages Used

Rbowtie2 for bowtie2() and bowtie2_build(); Rsubread::align() for BWA-style alignment; Rsamtools for BAM manipulation, indexing, and flagstat-style statistics; GenomicAlignments for downstream alignment-aware analysis.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat